

V2X communication and QUIC: A Comprehensive Review

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Abstract

Autonomous vehicles (AVs) rely heavily on Vehicle-to-Everything (V2X) communication to achieve safe, efficient, and reliable operation in dynamic traffic environments. While existing V2X deployments are primarily built upon DSRC, C-V2X, and 5G-NR at the physical and network layers, the choice of transport layer and application layer protocols remains an open research problem. Traditional transport protocols such as TCP and UDP exhibit limitations in highly dynamic vehicular networks, including poor support for connection migration, multiplexing, and low-latency data delivery. QUIC, a UDP-based secure and multiplexed transport protocol standardized by the IETF, and its extension, Media over QUIC, has emerged as promising alternatives, offering resilience to network changes, reduced connection setup latency and media streaming capabilities. While the QUIC protocol has already seen applications in some of the use cases in the industry, Media over QUIC has not yet been used in large scale.

This paper presents a comprehensive review of V2X communication technologies with a particular focus on QUIC and emerging protocols, especially Media over QUIC (MoQ). We survey the current state of V2X communication architectures, commonly used IoT application-layer protocols, and recent studies applying QUIC and its variants in vehicular environments. Furthermore, we examine the design principles, architecture, and current research status of Media over QUIC, highlighting its potential to support high-throughput, low-latency streaming of sensor and media data in V2X scenarios.

In addition, this review summarizes existing V2X simulation frameworks, including SUMO, CARLA, OMNeT++, and Veins, which enable systematic evaluation of communication protocols under realistic traffic and network conditions. Based on the surveyed literature, this review identifies key research challenges and open questions related to the adoption of Media over QUIC in V2X communications, including performance trade-offs, protocol interoperability, and suitability across different V2X use cases. This work aims to provide a structured foundation for future research on applying Media over QUIC to enhance the performance and reliability of next-generation V2X systems.

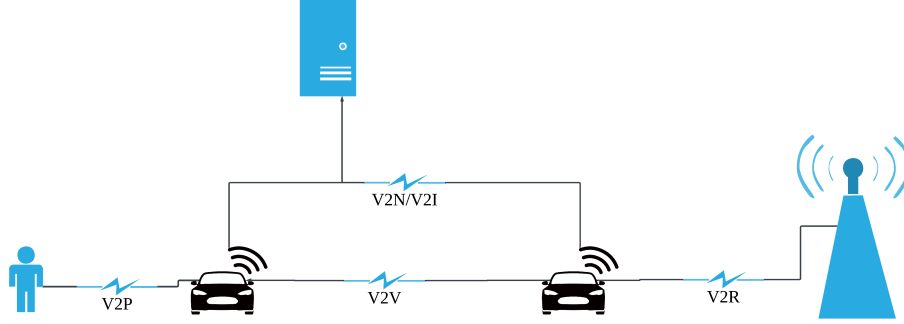


Figure 1: V2X Communications

1 Introduction

1.1 V2X communication and performance issues

Vehicles, such as cars, trains or planes, used to be solely operated by human drivers or pilots, however recently Autonomous vehicles(AV) have seen rapid development. The concept of AV is not a new invention, it was proposed as early as 1918[1]. However, with the relatively recent development of the Internet of Things (IoT) and Artificial Intelligence (AI), a network of autonomous vehicles could become feasible and see limited commercial use, both in enclosed space like airport or open urban roads[1]. In theory, a vehicle can be considered autonomous when all the tasks related to driving at all environment could be done by the vehicle's own system without human intervention[2]. With that in consideration, it's not difficult to draw the conclusion that communication between vehicle to everything(V2X) is vital for AV systems to operate without issues or accidents. V2X communications encompasses Vehicle-to-vehicle (V2V), Vehicle-to-Network (V2N) or Vehicle-to-Infrastructure (V2I), Vehicle-to-Road Side Unit (V2R), and Vehicle-to-Pedestrian (V2P).[3] There are many use cases for V2X communication, such as do not pass warning, forward collision warning, queue warning, parking discovery, optimal speed advisory, curve speed warning and fleet management.[3] With these use cases, V2X communication is expected to bring many benefits such as a drastic decrease in traffic-related fatalities, reduced logistical costs for operating vehicular fleets, and the introduction of a variety of new business models, for that the V2X communication has attracted a lot of attention.[4]

The Architecture of V2X networks is composed of multiple layers and functional entities to allow communications and vehicles. The decentralized three-layered architecture, composed of Edge, Fog and Cloud is considered to be one of the most adopted architecture, and this architecture can be utilized to support most of the requirements of V2X applications. The Edge layer is the physical layer, composed of entities and actors in the traffic, such as sensors, road signs, vehicles and pedestrians. The Fog layer is an intermediate layer between the

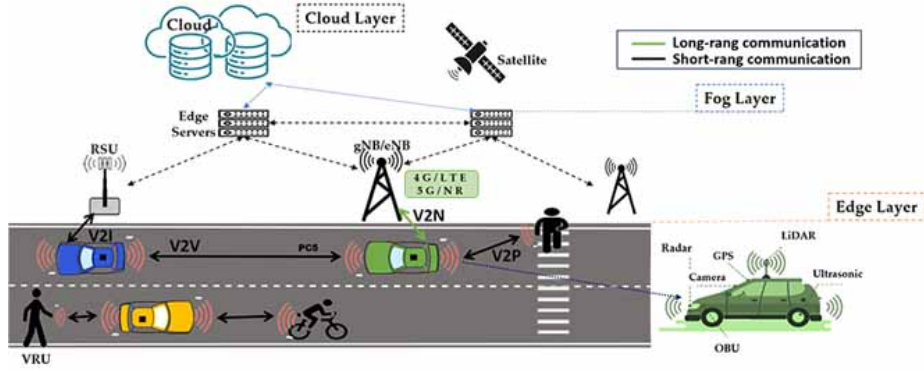


Figure 2: Image of V2X three-layer architecture[5]

Edge and Cloud layers, it is composed of network hardware such as Road-Side Units(RSU) or cellular base stations. The Cloud layer handles processing and storage of data collected from the Edge layer, and in turn provides service to the Edge layer such as providing traffic jam predictions. [5]

As the development of autonomous vehicles moving on from simply moving individual vehicle to putting the vehicle into use, V2X (Vehicle to everything) communication becomes vital. V2X communication technologies can be categorized by the distance between the vehicle and the receivers, in short distance, there's Bluetooth, ZigBee and UWB(Ultra wideband); in medium distance, the DSRC and Wi-Fi and in long distance the C-V2X and 5G-NR.[6] The recent trends shows that the C-V2X has become the main standards of V2X communication with the deployment of the 5G networks, with DSRC still being used in short to medium distance communications when network service is an issue.[5] The requirements for V2X communication can be categorized as safety, transport efficiency and information/entertainment applications,[7] and recently V2X is being integrated with AI and Machine/Deep Learning approaches to further enhance its capabilities.

For the transport layer above, it is more flexible and open to diverse implementations. Currently existing IoT application layer protocols are mostly built on TCP and UDP, they have suboptimal perform in a dynamic V2X environments[8], for example, TCP and UDP does not support multi-homing and multi-streaming, this leads to disconnection when network changes, which happens frequently in a V2X scenario. [8]

In use cases such as valet parking and platooning[9]. The application layer studies mentioned a requirement for a more efficient communication technology. A robust end-to-end communication with high data throughput and low latency to both increase the quality of the communication and open up new opportunities for implementing use cases that are considered difficult before, such as streaming media data to devices for better analysis.

To solve these issues, new protocols such as QUIC are employed. There are many combinations of IoT application layer protocols with QUIC that optimizes performance in the V2X communication, such as MQTT and WebSocket, and they have been tested by studies[8]. Recently, there is a new protocol emerging called the Media over QUIC (MoQ), this protocol is mainly used to stream media in a bidirectional way, and can be used in use cases such as video conferencing[10]. However, this protocol can also be used for things like P2P chat and other general information transmission tasks, and also have the potential to be used in V2X communications. The purpose of this review is to explore the state of the art in V2X communication, QUIC protocol, Media over QUIC, its potential application in V2X communication and also related testing and simulating methods. The goal is to pave the way to boost the performance and reliability of V2X communications by applying Media over QUIC and testing the metrics.

1.2 QUIC Protocol

QUIC protocol is a multiplexed and secured protocol based on UDP standardized by the Internet Engineering Task Force (IETF) in RFC 9000. It was first implemented and deployed in 2012, and it is a protocol that creates a stateful connection between a client and a server.[11]

The QUIC protocol integrated TLS encryption within the packet, making it more secure compared with TCP and UDP, and by integrating the TLS handshake with QUIC handshake, it permits exchange of application data as soon as possible, an advantage over TCP+TLS, which requires another handshake.[11] QUIC packets include QUIC frames, which carry control information(headers) and application data between clients and servers, and authentication is also included. QUIC packets are carried in UDP datagrams to both utilize existing networks and increase performance. [11]

Many applications are built over QUIC today, as it already replaced TCP in HTTP/3 protocol because of its stream multiplexing, TLS integration and flow control abilities and improved HTTP performance.[12] Currently, there are several other protocols built upon QUIC to exploit the functions it provides, such as MQTT over QUIC, and Media over QUIC. MQTT over QUIC already saw usage in the field of Internet of Vehicles.[13]

1.3 Media Over QUIC

Media over QUIC Transport (MOQT) is a relatively new protocol, its purpose is to stream media, like video, over QUIC. The MOQT standard is currently still a draft and subject to change[14], however there already are beta implementations of this protocol in Go, Rust, TypeScript and other languages. The motivation behind the MoQ is to reduce latency, to deliver content in a higher bitrate and to reuse some of the functions provided by QUIC, such as congestion control and streaming. Also, the MOQT intends to develop a single protocol which can be used for both media ingest and distribution[14]. The MOQT also intends

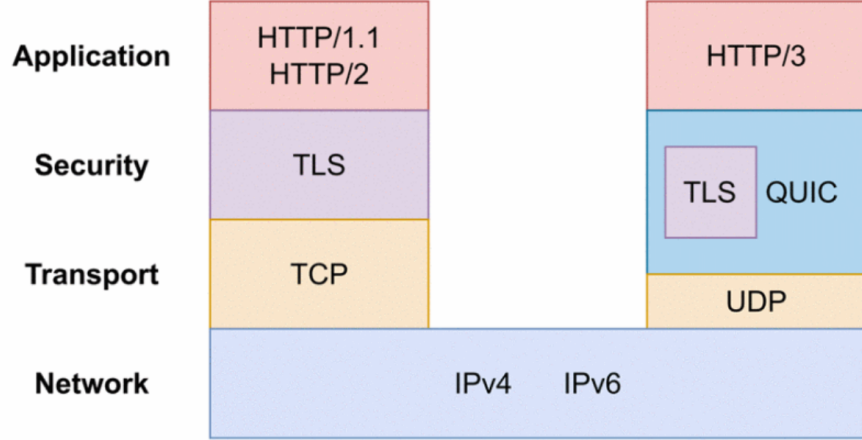


Figure 3: Network protocol stack comparison, highlighting the evolution from HTTP/1.1 to HTTP/3[13]

to implement a relay service that grants greater scalability by having the relay service cache content for distribution and simultaneously routing content and responding to congestion[14].

Media over QUIC is created to distribute media contents such as video, but in reality it can be used to distribute any binary data flow. The MOQ data flow is organized in groups and groups are organized in tracks. A track is like a topic in a publish/subscribe message queue, the consumers can subscribe to a topic and receive the data flow from this topic[14].

As shown in figure 3, MOQ is composed of 4 layers in its architecture, the underlying layer is QUIC or WebTransport, as a failover protocol. This layer is responsible for data transmission and providing the functions provided by QUIC. Above is the moq-lite, this is the implementation of Media over QUIC, and provides relay service to allow the endpoints to subscribe and publish tracks. Above is the media processing layer called Hang, this layer is in charge of media related tasks such as encoding and decoding and media streaming. Above that is the application layer where business logic is being implemented. If no media is being transmitted, the media layer can be omitted and data can be send between application layer and MOQ layer directly.

1.4 IoT protocols in V2X

IoT, or the Internet of Things, is the technology that connects devices in the real world together through a common network, it is often based on wireless connection, such as Bluetooth, Wi-Fi or cellular network. This is similar to

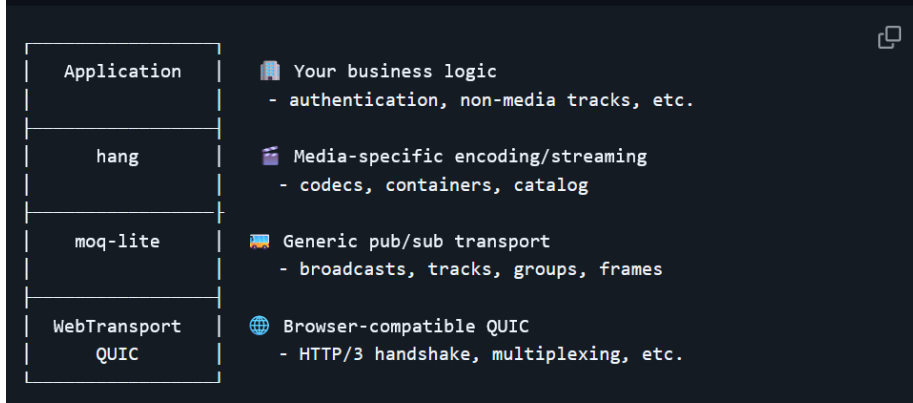


Figure 4: MOQ library architecture[15]

Levels	Function	Human Driver Participation
L0-L2	Escort Driver	Driving is done by human, system is only assisting the driver
L3	Substitute Driver	Driving can be done automatically in some cases
L4-L5	Exclusive Driver	Driving can be mostly automated in all situations

Table 1: Levels of autonomous driving[17]

V2X communication, as V2X is also connecting vehicles with other devices, V2X communication is also a form of IoT, or IoV(Internet of Vehicles). IoV is a dynamic network infrastructure connecting vehicles, users, and other smart devices to the Internet[16]. The evolution of 5G and 6G cellular network enabled vehicles to share real-time data, achieve cooperative perception, and collaborative decision-making, this significantly advancing the implementation of Level 4/Level 5 (L4/L5) autonomous driving[17].

In application layer of the OSI seven-layer model, the IoV utilized a wide range of protocols to allow vehicles to communicate with each other and other devices. To achieve effective communication in a vehicular setting, a secured protocol with low bandwidth and energy consumption is desired. Study finds that Message Queuing Telemetry Transport (MQTT) and Constrained Application Protocol (CoAP) are often used in this scenario[18]. Although they also have several issues that requires solving. The MQTT is a binary protocol and normally requires a fixed header of 2 bytes with small messages up to 256 MB in size, and it works in a pub/sub model. It can use both TCP+TLS and QUIC as it's base protocol in order to achieve quality of service requirements. Due to size constraints of each MQTT packet, it's not efficient to stream media or large binary files using the MQTT protocol. CoAP supports both request/response and resource/observe (a variant of publish/subscribe) architecture. REST architecture is used as a communication between HTTPclient and server because of its lightweight-ness and is easier to consume. CoAP is built upon UDP and

uses the Efficient XML Interchanges (EXI) format to transmit the messages, and CoAP also allows both reliable and unreliable messages to allow different application when needed. However, it is less reliable than MQTT due to it being based on UDP[19]. Both MQTT and CoAP have some issues in its application in the IoV area. The MQTT requires a persistent connection, and is inefficient when streaming media[20], both limited its performance in use cases such as fast-moving vehicles and streaming sensing data to other vehicles and devices. CoAP has limited scalability in some cases, making it inefficient in a large scale network of vehicles. There are some works that uses MQTT in a real-time media processing scenario, but MQTT is only used to send processing results in texts instead of the media itself, limiting the processing only on the local machine.[21]

Challenges in IoV includes interoperability of network architecture, real-time massive data processing, intelligent routing and path planning and sensors and artificial intelligence integration[22]. In real-time data processing, a parallel data acquisition, data processing and big data analytics are needed and are essential in IoV. For data acquisition of massive real-time data, it's difficult for current protocols to satisfy this requirement.

1.5 V2X Simulation

It is often difficult to test V2X communication in real vehicle environments, due to material and funding constraints. In related studies, simulation is considered to be the ideal approach of creating a good approximate of the traffic and network environment as they would perform in the real world. There are two main challenges in simulating a V2X environment, first is to simulate a traffic environment, and second is to simulate a network environment, especially the simulation of cellular networks when the vehicle is moving fast and are frequently changing the cellular base stations it connects to.

CARLA is an open source autonomous driving simulator that can support development, training, and validation of autonomous driving systems. In addition, it also provided a detailed virtual scene in Unreal Engine, allowing detailed, physics-based simulation. It can simulate sensors, vehicles and road-side unit actors and have them interact with each other in the scene. Carla can also integrate with SUMO for further enhanced simulation features, such as NPCs controlled by SUMO. CARLA is scalable, flexible and allows easy customization, this makes it suitable to build an ecosystem of simulators based on CARLA.[23]

OMNeT++ is an extensible, modular, component-based C++ simulation library and framework, primarily for building network simulators. It can simulate both wired and wireless networks, and it supports domain-specific simulations such as sensor networks, wireless ad-hoc networks, Internet protocols, performance modelling, photonic networks, etc., and there are also extensions for real-time simulation and network emulation.[24]

Veins is an open source framework for running vehicular network simulations. It is based on two other simulators: OMNeT++ and SUMO. It extends these to offer a comprehensive suite of models for IVC simulation. Veins allows for online

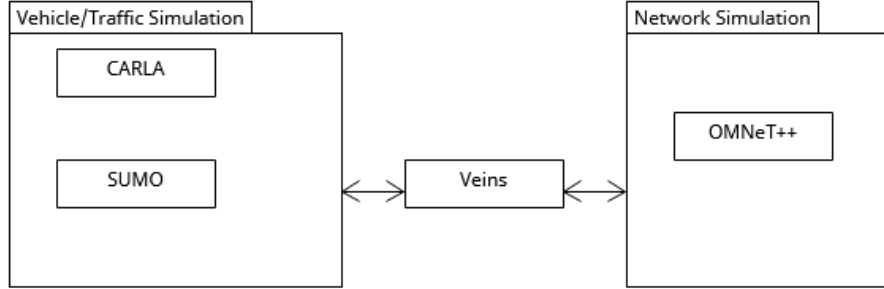


Figure 5: Simulation system architecture

Tool	Role
CARLA	Simulates autonomous driving scenarios
Veins	Combines network simulation (OMNeT++) with road traffic simulation (SUMO)
PTV Vissim	Incorporating AI for traffic simulation in a micro scale
V2XSim	Simulates connected and automated vehicle environment simulations

Table 2: Tools used in study conducted by Panyam et al.[26]

re-configuration and re-routing of vehicles in reaction to network packets. It relies on trusted vehicular mobility model and implementation of human-driven and/or autonomous cars. Veins relies on models of IEEE 802.11p and IEEE 1609.4 DSRC/WAVE network layers, and can include third-party models such as 4G/5G cellular networking like 3gpp LTE and C-V2X directly or via the INET Framework, making it versatile in simulating vehicle networks.[25]

2 Literature on QUIC Protocol

Current studies on QUIC focused on QUIC performance and its comparison to existing protocols like TCP and UDP, especially the comparison between QUIC and TCP+TLS.[27] A search on Google scholar with related keywords yielded a significant number of results, as shown in table 2.

2.1 QUIC VS TCP

A study conducted by Shreedhar et al.[28], evaluated QUIC performance over web, cloud storage and video workloads and compares them with traditional TCP+TLS. They observed that QUIC has up to about 140% lower mean connection times than TLS over TCP for low-bandwidth and high-RTT networks, and they found out that IETF QUIC performs better than different versions of gQUIC, and for video workloads, they find that QUIC is 400-500 ms faster than TCP+TLS in establishing a connection to YouTube media servers. For video workloads, they concluded that QUIC tends to depict better video content de-

livery with reduced stall events and lower stall durations of up to 50% due to its lower latency overheads.

Another study conducted by Jeddou et al.[29], compared TCP with QUIC over mmWave links used by 5G cellular networks. The results show that QUIC outperforms TCP in highly fluctuating mmWave channels, showing better performance for both throughput and latency. In addition, a study conducted by Peng et al.[30], evaluated TCP and QUIC performance on cloud control systems. The result show that when a control system operates at a relatively low frequency and network latency is not significant, QUIC exhibit worse performance than TCP. However, under higher frequency with significant network latency, QUIC demonstrates shorter and less variant response delay, which is beneficial for more time-critical applications and advanced control designs.

A study conducted by Qin et al.[31], also compared TCP and QUIC, and their conclusion is QUIC outperforms TCP in most experiments for web browsing and online video, but its performance is susceptible to congestion control algorithms and network conditions. For example, when using the Cubic algorithm, QUIC enabling the 0-RTT feature can decrease the webpages loading time by 37.11% compared with TCP. But when using the BBR algorithm, both QUIC and TCP achieve high throughput, slight fluctuation, and few delayed events when playing online videos. Based on these studies, we can say that in scenarios where high throughput and low latency is desired, QUIC outperforms TCP and TLS, but in other scenarios, QUIC performance is equivalent to the level of TCP.

2.2 QUIC in V2X

When applied to V2X communications, QUIC also displays advantage. In a study conducted by Suleman et al.[8], they evaluated ALPs including AMQP, CoAP, DDS, MQTT, WebSocket (WS), and XMPP, over QUIC, SCTP, UDP and TCP. They measured metrics like connection establishment time, latency, PDR, throughput, and packet inter-arrival time, and their conclusion is that CoAP over QUIC/UDP and WS over QUIC demonstrated significant potential for transmitting messages across diverse V2I scenarios, especially when network congestion intensifies.

A study conducted by Lee et al.[32], compared TCP and QUIC performance with congestion control algorithms (CUBIC and BBR) over V2V Wi-Fi. In their study they found out that TCP with BBR delivers the highest performance, achieving up to 250 Mbps at intersections, however, the result is also influenced by relative distance, speed and Wi-Fi signal strength.

Another study conducted by Suleman et al.[33], studied data quality assurance performance of different protocols in V2I networks. The testing they did shows that in the highly dynamic network of V2I, a singular IP-based protocol is difficult to support all the data quality requirements, and a fusion between different protocols and different layers are needed for a efficient V2I network.

Keywords	Search Result Counts
"QUIC" + "Performance"	27000
"QUIC" + "Performance" + "TCP"	7360
"QUIC" + "Performance" + "UDP"	6100

Table 3: QUIC research keyword results on Google Scholar

2.3 Downside of QUIC and Discussion

However, there are some downsides of using QUIC as well. The study conducted by Oran et al.[27], show that while QUIC is faster than TCP, the QUIC protocol uses more processing power, for example, the encryption overhead of the QUIC protocol is high. The QUIC protocol encrypts packets with TLS. Therefore, the use of the QUIC protocol may not be appropriate in scenarios where CPU usage is low, which might become a bottleneck in certain cases, such as devices with more energy or processing power constraints.

In conclusion, QUIC protocol is very promising and is applicable in the V2X networks, and there are already works using them in V2X communication. However, there are certain issues with default QUIC such as higher energy and processing power consumptions and congestion control algorithms. It is advised to use a modified version of QUIC and combine it with other protocols to achieve optimal performance.

3 Literature on Media over QUIC Transport

Media over QUIC is still relatively new and the standard is still in the draft phase, the studies related to MOQ are focused on functionality and performance testing.

3.1 MOQ Performance

The study conducted by Gurel et al.[10], studied one of the initial proposals in detail, and did a gap analysis. They tested using the Warp protocol, an implementation of the MOQ. They extended Warp and implemented two algorithms for bandwidth measurement and it shows that the result is the benchmark provided by their algorithm perform better than the server-side estimated throughput(ETP), meaning MOQ compares better than what current servers could measure.

Another study conducted by Gurel et al.[34], compared between MOQ and other media protocols such as low-latency DASH(LL-Dash), and the result shows MOQ performs better than LL-Dash in terms of latency, stall duration and bandwidth.

Keywords	Search Result Counts
"V2X" + "Performance"	70100
"V2X" + "Integration"	58900
"V2X" + "TCP"	7180
"V2X" + "MQTT"	2000
"V2X" + "QUIC"	370

Table 4: V2X research keyword results on Google Scholar

3.2 MOQ Application and challenges

A study conducted by Engelbart et al.[35], discussed MOQ application in DNS. They explored a publish-subscribe variant of DNS based on the Media-over-QUIC architecture, and they find that DNS can benefit from this publish-subscribe variant by not only limiting update traffic, it can also considerably reduce the time consumed for a resolver to receive the latest record, thereby supporting use cases, such as load balancing in CDNs. However, it also brings new challenges to the DNS, including a higher overhead for endpoints due to state management, and increased query latencies on first lookup caused by session establishment latencies.

However, MOQ is still a new protocol and it faces some challenges, such as standardization, security and interoperability with other protocols, and also increase data transmission efficiency for small and frequent packets. An IETF draft by Yue et al.[36], mentioned that MOQ must resolve issues such as reducing overhead for small and frequent data packets, optimizing transmission for bursty data patterns, and ensuring low-latency delivery when under high traffic loads.

3.3 MOQ Prospect

Existing studies shows that MOQ is promising and it can achieve better performance compared to some of the already optimized media streaming protocols. It has already gained a lot of attention from industries and Cloudflare has released the first MOQ CDN[37]. MOQ can be utilized in many areas to achieve higher throughput and lower latencies, improving performance and user experiences.

4 Literature on V2X Communication

The current studies for V2X communication applications focuses on system integration of V2X with other protocols and technologies and their performance. For integration, studies focus on integration with IoT application layer protocols such as MQTT. For performance, the studies focus on performance of various underlying protocols for V2X, such as C-V2X, DSRC and 5G-NR, and also how transmission layer and application layer protocols perform in V2X scenario. Table 3 shows the distribution of integration of V2X and other protocols and V2X performance in the current landscape.

4.1 V2X Performance

In a study conducted by Tahir et al.[38], they compared the performance of TCP and UDP in a V2V/V2I communication scenario based on IEEE 802.11p and 5G using field testing methods. From their experiment, they concluded that UDP is preferred over TCP for its lower latency and higher throughput, although UDP have issues with packet losses and lower reliability.

Another study conducted by Khatiwoda et al.[39], surveyed how different IoT ALPs, namely MQTT and ZeroMQ, adapt to low-latency requirements in 5G V2X communications. They found out that ZeroMQ outperforms MQTT in terms of latency due to ZeroMQ's lightweight and brokerless architecture.

Based on these studies, we can see that in V2X communications, low latency is a critical requirement and need to be satisfied from transport layer to application layer, and requires a fusion of multiple protocols and technologies to achieve that.

4.2 Integration and Application of V2X and Other Protocols

There are studies such as the one conducted by Huang et al.[40] that studies bandwidth pressures brought by video streaming requirements in vehicular networks, and demonstrated streaming of video between vehicles using experimental 6G network features, and proposed a method that integrates millimeter-wave (mmWave), terahertz (THz) frequency band, and cell-free (CF) extremely large-scale multiple-input multiple-output (XL-MIMO) wireless communication technologies to implement this feature. In the study, they listed current issues in the approach they use are the deployment cost and energy consumption is high, difficulty in ensuring communication stability due to low coverage distance of the antennas, and moving vehicles often leads to change network topology, causing connection to be disrupted.

In application layer, there are explorations of using QUIC, unreliable QUIC and MPQUIC(Multi-Path QUIC) for video streaming in V2X communications in urban scenario[41], and they reach the conclusion that unreliable and multi-path protocols provides more advantage in low-latency and network robustness in vehicular networks.

In the study conducted by Amponis et al.[42]., they surveyed the application of QUIC in enabling low latency and resource aware vehicular communications and the issue of QUIC's low performance in a highly dynamic vehicular network due to QUIC's default congestion and flow control algorithms. They proposed a new algorithm and optimized its performance in V2V networks, and concluded that customized protocols performs better in V2X scenarios than default protocols.

In conclusion, in V2X communications, low latency and high throughput requirements are vital, and protocols must be carefully chosen and customized to achieve these requirements in a highly dynamic network. The best option, based on the studies above, is using a customized, unreliable and multi-path

QUIC-based protocol(that in itself is based on UDP) in a V2X network as a transport layer protocol, instead of pure TCP, UDP or QUIC with default settings.

5 Literature on V2X Simulation

5.1 Application of V2X simulation tools

Recent studies regarding V2X simulation listed a large array of methods and tools of simulating both traffic data and network environment. In a study of using AI and LLM in V2X communication[26], the researchers used SUMO (Simulation of Urban Mobility) to simulate traffic and integrated AI and machine learning with it. The simulation framework included CARLA, Veins, PTV Vissim and V2XSim.

In a study compared by Bernad et al.[43], they surveyed the options of selecting a 5G simulation tool for performance analysis of smart edge computing orchestration algorithms in a urban environment for vehicular/smart cities network. They compared 5G simulation tools like Simu5G and 5G-Sim-V2I/N, both built as extensions of OMNeT++, and they found out that they perform differently even with same input, and the selection of 5G simulation tools can greatly impact the conclusion of the research. In the 5G-Sim-V2I/N vs Simu5G comparison, the Simu5G framework turns out to be the clear winner with behaviour and results within the expected ranges according to the 5G specifications.

A study conducted by Guo et al.[44], proposed a method to automatically generate test scenes for cooperative perception in V2X called V2XGen. The experimental results show that V2XGen is capable of generating realistic test scenes and effectively detecting erroneous behaviors in different V2X-oriented driving conditions. The results also validate that retraining systems under test with the generated scenes can enhance average detection precision while reducing occlusion and long-range perception errors.

5.2 Comparison of simulation tools

A review on V2X cybersecurity integration in V2X extensively reviewed a large number of V2X simulation tools[45]. From analysis of the literature they reviewed, SUMO, OMNeT++, and Veins are the most frequent combinations of tools in the area of V2X simulation, because they offers complementary capabilities to each other and satisfies requirements in V2X research, with SUMO providing traffic simulation, OMNeT++ provides network simulation and Veins acts as a bridge between them.

In a review conducted by Ashayer et al.[46], the author addressed the crucial role of co-simulation in advancing ntelligent transportation systems(ITS) and connected and autonomous vehicles (CAV) technologies, highlighted the limitations of single-platform simulations in capturing the benefits of integrated

simulation environments. They reached the conclusion that a singular simulation tool is hard to simulate the complexity of urban traffic and the network situation within the traffic, and aco-simulation tools would be required to simulate such a scenario.

In conclusion, to simulate the complex environment in a V2X network, a suite of simulation tools is required, as a singular tool can't satisfy all requirements such as traffic simulation and network simulation, a co-simulation approach is essential in V2X studies. Most commonly used tools are SUMO, Veins and OMNeT++ to provide both traffic and network simulations, while tools like CARLA and V2XSim can enhance the capabilities of traffic simulation. In terms of 5G simulation, Simu5G is the optimal choice. Also, It is important to create a suitable testing scenario to conduct accurate and adequate testing, and a series of automated tools can be employed for this task.

6 Conclusion and Discussion

V2X communication is a rapidly growing area as it provides many benefit in autonomous vehicles, improving road safety and improving user experience. The application of a better, more efficient application layer protocol is needed by autonomous vehicle developers to implement advanced features and achieve L4-L5 autonomous driving. To allow efficient and secure data transmission, current IoT application layer protocols such MQTT and CoAP are applied to send telemetry data between vehicles and sensors.

QUIC protocols are now being widely used as it combines some of the useful features, like congestion control, of TCP with hi-speed communication provided by underlying UDP. QUIC also provides streaming and multiplexing, allowing connection between endpoints to recover after network topology changes. Applying QUIC in V2X has many benefits, the streaming and multiplexing features allows for timely service restoration after network topology changes, which happens frequently in V2X scenario.

Media over QUIC is a emerging protocol that allows streaming media content, such as video, over QUIC. This fits well with the requirement of V2X communication, as V2X often requires streaming sensor data with other vehicle and devices, and the sensor data might be large in size. Currently in V2X communication, the sender vehicle processes the information it received and then send it via other protocols to other devices, but it is desirable to send raw sensor data to a processing service, to save processing resources on the vehicle and therefore reducing energy consumption on the vehicle.

The simulation tools also have developed significantly. A mature combination of tools like SUMO, CARLA, Veins, OMNet++ and V2XSim allows simulating traffic scenario and network conditions for vehicles and devices in the V2X network. With these simulation tools, researchers can simulate various V2X scenarios easily before testing in a real device or vehicle, which could be cost intensive.

With all that in consideration. a new research direction can be proposed:

The application of Media over QUIC in the field of V2X communication, with performance benchmarking in various use cases and scenarios using simulation tools. The potential benefit is that MOQ can enable vehicles to stream data to other devices via both cellular network and Wi-Fi more efficiently, better increase road safety and implement better autonomous driving. However, several points should be further studied and considered:

RQ1: How can we design a system that integrates MOQ with V2X communication?

RQ2: Does MOQ perform better than MQTT, both over QUIC and over TCP, and general TCP and UDP in terms of data throughput and latency?

RQ3: What V2X use cases can be boosted by the application of MOQ, in terms of latency and data throughput?

As the V2X network expands further in size, its abilities would also grow. The current bottlenecks have become more and more significant and demands for a better protocol and better communication technologies grows every day. Future works can further explore the area of application of the MOQ, as for now there is not much studies regarding this area.

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